Inference-time proofreader guidance

Can the released proofreader improve complete contact maps by changing sampling, looking ahead, maintaining several prefixes, repairing errors, or combining larger independent rollout pools?

Frozen generator: exp277 step 266344. Frozen proofreader: exp356 step 18000.
Implementation and GPU validation are in progress; no guided-policy result yet.

Frozen models and experimental references

The generator is the exp277 step-266344 Qwen3 model. The proofreader freezes its causal backbone and adds a four-layer, 512-wide bidirectional contact reader with 22.18 million trainable parameters. It predicts each supplied contact's correctness and the current rollout's recall. No model is retrained here.

The proofreader was trained on 50,000 experimental PDB chain structures. References use the complete exp222 native-only pyconfind contact set: a 3 Å side-chain threshold, minimum contact degree 0.001 and sequence separation ≥6. The validation split has 1,552 structures in 287 related groups.

What top ten percent means here

At each contact boundary, sample K candidate contacts from the constrained generator. Score them with the proofreader after appending each candidate. Keep the highest ceil(0.1 K) candidates and sample uniformly among survivors. This keeps 2 of 16 or 7 of 64 sampled proposals, including possible repeats; it does not enumerate the entire residue-pair universe.

Soft guidance weights the same proposals by q raised to a chosen power. Lookahead adds future contacts before scoring, then commits only the first. The generator's probability already enters through proposal sampling.

Complete nested-pool pilot

On 32 validation structures, sum-q R-precision increases from 0.5792 with eight rollouts to 0.5993 with 128. The paired gain is 0.0200, with 95% group-bootstrap interval 0.0050–0.0329. Long-range R-precision rises from 0.5537 to 0.5806.

Candidate coverage grows from 0.7765 to 0.9509. At 128 rollouts, frequency alone slightly exceeds sum-q: all-contact R-precision 0.6040, long-range 0.5864. Most remaining error is ranking error despite the much larger candidate pool.

Confidence tie-breaking and predicted-F1 vote weighting were added as exploratory validation tests. Their tiny gains over frequency have intervals including zero.

Baseline calibration and coverage

On 128 ordinary rollouts per pilot structure, per-rollout predicted precision has mean absolute error 0.0606 and predicted recall 0.0547. Errors average within structures and then over the same 32 structures. Contact calibration curves weight individual contacts; the two summaries answer different questions.

Mean trajectory overlap is 0.3473. Enlarging the pool exposes about 95% of native contacts, but ranking only recovers about 60% at the native contact-count budget. The remaining ranking gap motivates guidance and better aggregation; it does not establish that either can close the gap. Guided-rollout calibration is pending.

Practical single-batch baseline

Requesting each pool in one batch gives 0.6036 all-contact and 0.5894 long-range R-precision with frequency ranking at 128 rollouts. Mean warm generation cost is 7.165 H100 seconds per map; no proofreader pass is needed for this aggregator.

Against eight rollouts with sum-q ranking, the paired all-contact gain is 0.0244 (95% group interval 0.0101–0.0373) and the long-range gain is 0.0357 (0.0154–0.0519). The 128-rollout pool contains 95.39% of native contacts. The single-batch recipe is the practical cost baseline for guided search.

Exploratory length control

On the five short proteins completed by every original policy, top-10% filtering raises per-rollout precision but emits fewer contacts and lowers recall. This incomplete cohort motivates a stopping-rule ablation, not policy selection.

An added exploratory pilot compares ordinary and top-10%-of-16 sampling at exactly L and 2L contacts, with EOS disabled for both. Four independent 2L continuations yield nested L and 2L maps. The original 15-policy pilot remains frozen. No new reserved-test scores have been inspected.

Existing 1552-structure validation baseline

On existing validation rollouts, sum q remains strongest among five simple aggregators: all-contact R-precision 0.5315; long-range R-precision 0.5046. Frequency gives 0.5167 and 0.4877 respectively. Mean q alone gives 0.4895.

With up to eight rollouts, the pool contains about 75% of native contacts. This separates missing-contact generation from ranking errors. Oracle pool coverage is an upper bound, not the performance of an inference method.

Frozen pilot and execution

32 validation proteins from distinct homology groups span 33–937 residues. Compare top-percentile gates, soft weighting, lookahead, population search and one-pass repair, plus ordinary vLLM pools of 8, 32 and 128 rollouts.

Measure full-universe AP, all/long-range R-precision and complete search cost. Select policies on validation, then lock a reserved-test confirmation plan. Jobs use co-located CoreWeave H100s and resumable per-rollout artifacts.

Execution recovery

An eight-H100 RNO worker was preempted repeatedly. Iris failure-retry limits do not limit preemption retries, so the worker repeated its shared model transfer and lost local progress. It was canceled on detection; the incident is recorded.

Recovery uses original source snapshots and durable S3 trajectories. After East capacity remained full, a durable 6.014 GB RNO2A input mirror was completed with per-chunk reservations and a restart-safe 7 GB cap. Regional H100 smoke tests verify identical model and target hashes. The complete guided comparison is pending. Ordinary generation is nearly complete for 1,375 structures from 255 validation groups outside the pilot. New job starts are delayed by a regional controller outage. No new reserved-test scores have been computed or inspected.

Numerical validation

All frozen backbone tensors match the generator exactly. CPU cache tests agree with full forward passes. The first H100 smoke caught BF16 incremental/full-pass drift of up to 0.02394 in one contact probability. FP32 cached search passes on three real proteins, with maximum error 0.00000358. Final maps use the released BF16 full-pass reader for comparison.

Regional recovery status at 1730 UTC

East-region output writes failed after 16:55. All affected workers stopped, and 4,446 saved objects were copied to the GPU region. Source checksums and complete destination readbacks verified every file. Original code, sharding and seeds are preserved, with new results resuming from the verified regional snapshot.

The regional controller stopped responding after 17:14. Thirteen replacement and smoke jobs remain queued. Existing validation-generation workers continue writing outputs. These are infrastructure delays; guided-policy conclusions remain pending. No shared cluster state has been changed.

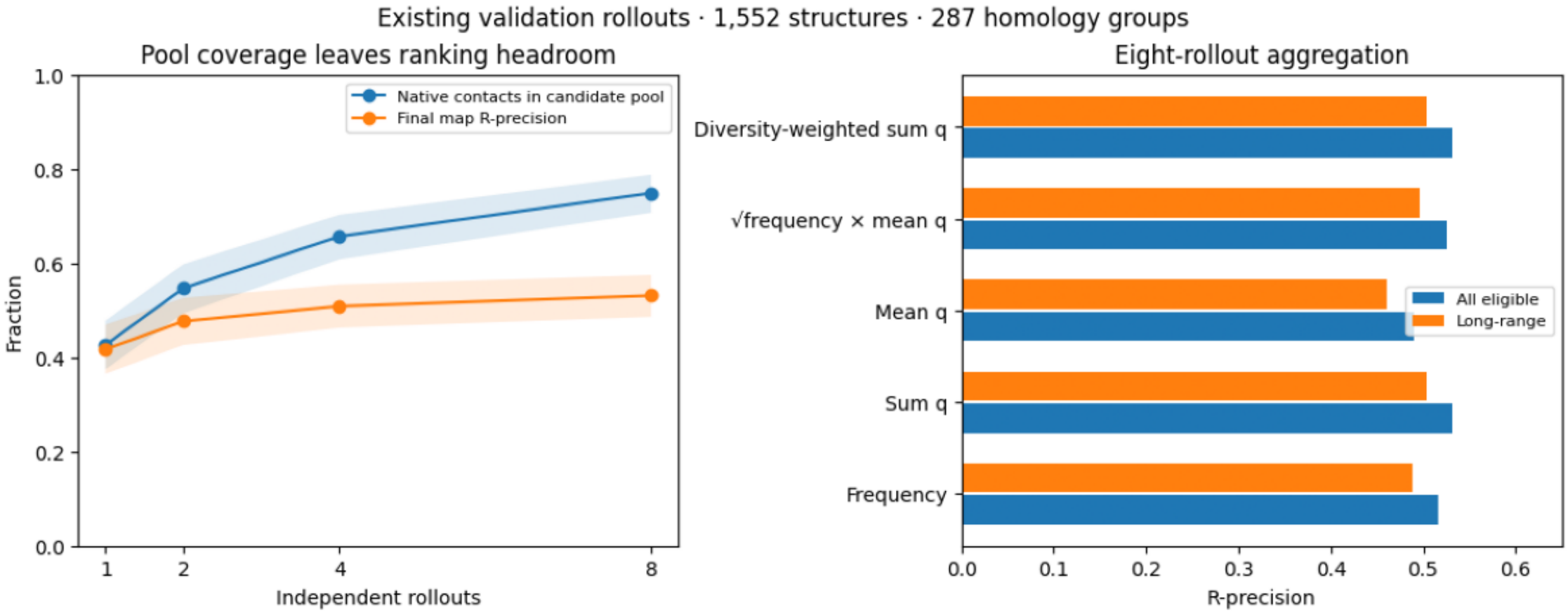
How to interpret the eventual comparison

R-precision evaluates the top R predictions, where R is the number of native contacts used only for evaluation. Precision at L, L/2 and L/5 provides usable sequence-length budgets. AP includes every eligible unproposed pair at score zero. Contact labels enter only after sampling decisions are finished.

Pilot and larger-validation results select the final methods. The reserved test was previously used for the proofreader release; new inference-policy test scores have not been inspected. Generator pretraining decontamination is not established. Search timings describe this implementation, including discarded work, and are not hardware-independent lower bounds on the cost of these policy families.

existing_validation.png

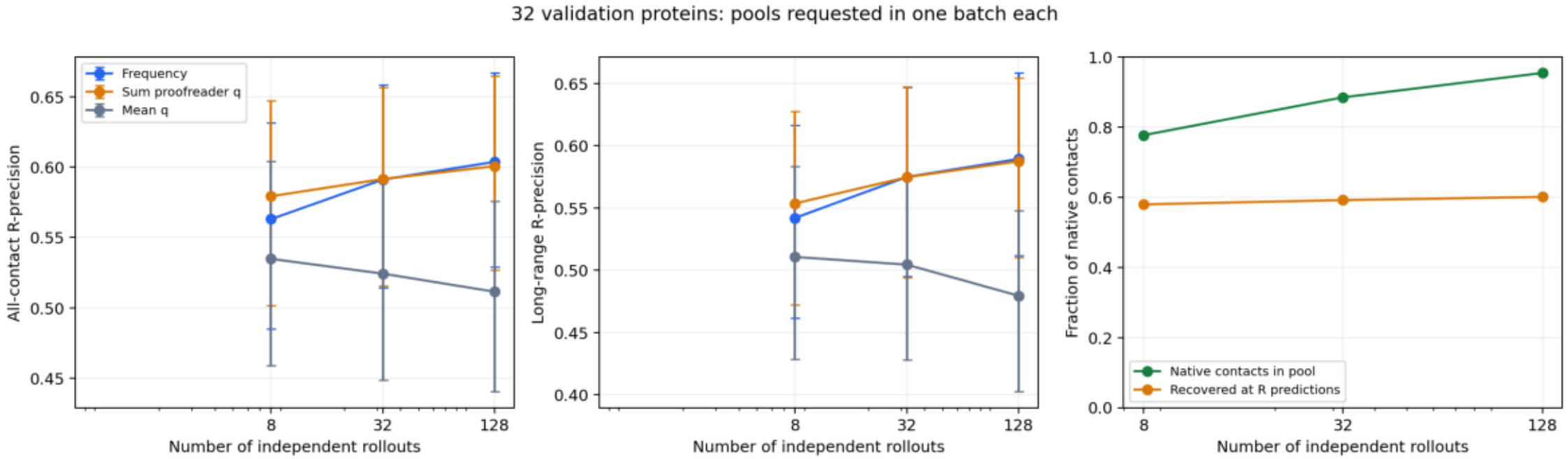
Validation-only reanalysis. Bands resample homology groups (95% intervals). Oracle coverage uses labels only to bound achievable recall. The small number of structures with fewer than eight usable rollouts retains all available samples. Long-range means separation ≥24.




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$ plot_existing.py
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pilot_direct_pool.png

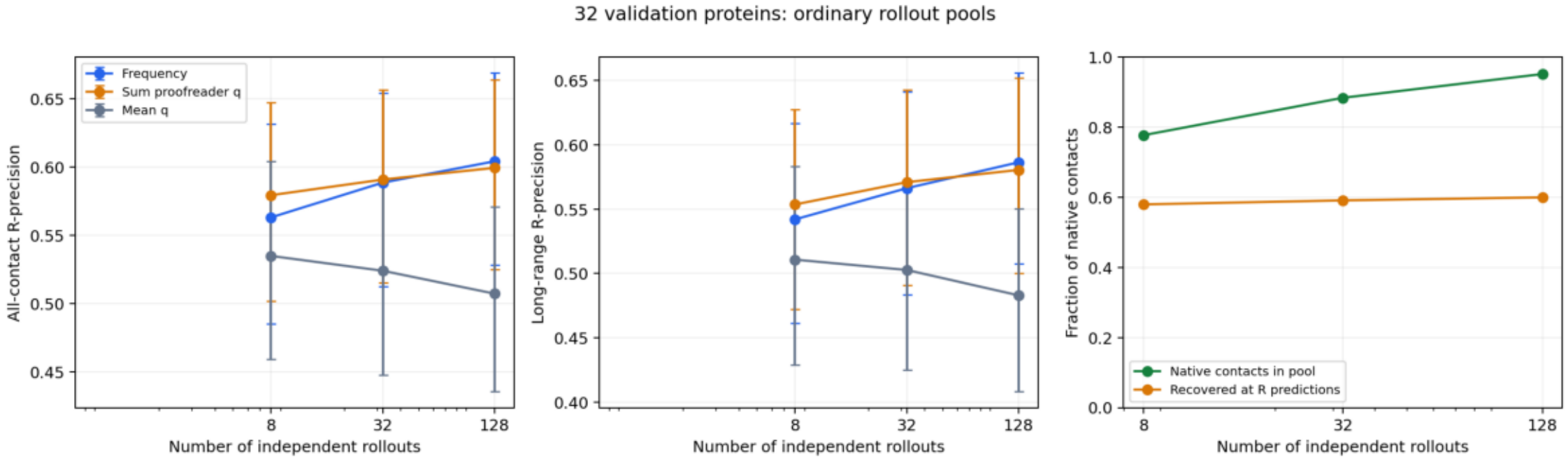
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_direct --out experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_direct_pool.png
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pilot_pool.png

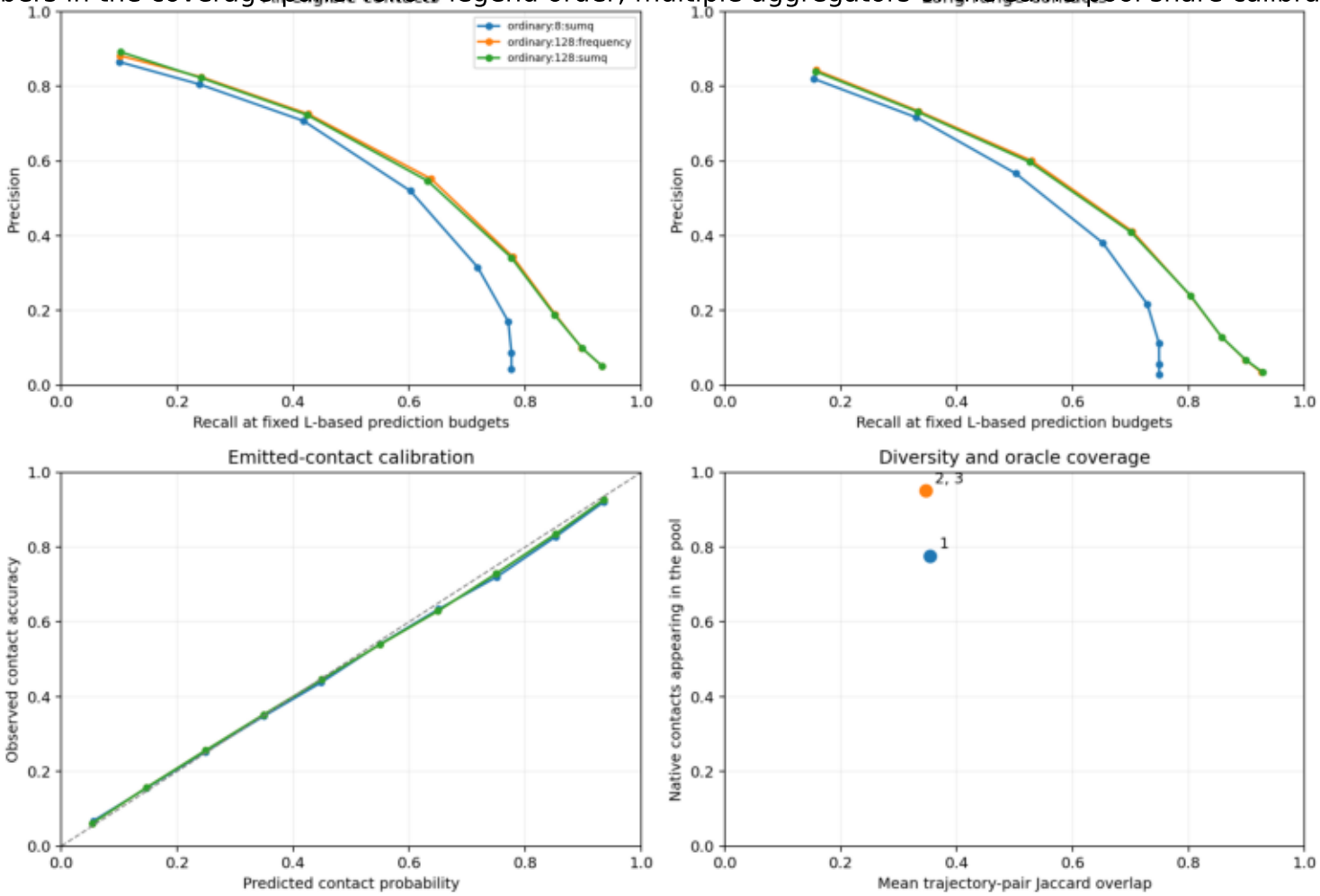
Mean over the same validation structures. Error bars are 95% homology-group bootstrap intervals. Oracle coverage measures whether a native contact was proposed; it is not an attainable ranking result. Long-range means sequence separation at least 24. No test-set results appear here.




```
$ plot_pools.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_pool --out experiments/exp368_evals_proofreader_guided_contact_maps/plots/pilot_pool.png
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pilot_pool_diagnostics.png

Curves average over the same structures at budgets 0.1, 0.25, 0.5, 1, 2, 4, 8 and 16 times sequence length. Unfilled slots count as misses; curves do not use native contact counts to choose the prediction budget. Calibration pools first occurrences within each rollout and is contact-weighted. Diversity and coverage are structure means, at most 128 deterministically sampled trajectory pairs measure overlap. Numbers in the coverage panel follow legend order; multiple aggregators of the same pool share calibration and diversity.




```
$ plot_diagnostics.py --data experiments/exp368_evals_proofreader_guided_contact_maps/data/pilot_pool/diagnostics --spec ordinary:8:sumq ordinary:128:frequency ordinary:128:sumq --title Ord
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